FINANDEX

FinanDex Whitepaper

Global Financial Market Indexing Insights — 27 data chatbots in one app

Version 1.0.0 · 21 October 2026

Finandex is designed and developed by Dea Saka Kurnia Putra.

A BRIGHTER TOMORROW THROUGH MEANINGFUL WORK

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Contents

01 What is FinanDex?

02 The Problem

03 How It Works

04 27 Dex in 8 Categories

05 Data & Methodology

06 The AI Layer

07 Technical Architecture

08 Security & Privacy

09 Platforms & Accessibility

10 Quality Assurance

11 Status & Limitations

12 Disclaimer & Copyright

CHAPTER 01

What is FinanDex?

FinanDex is a financial and socio-economic data insight app that brings **27 specialist chatbots ("Dex")** together in one place. Each Dex masters one topic — NASDAQ stocks, the rupiah, crypto, salaries, house prices or legal documents, for example — and answers your questions with numbers computed from real datasets, complete with charts, tables and an explanation of the method.

At a glance

- **One question, a data-backed answer.** Ask in plain language; Finandex AI routes you to the right Dex.
- **Numbers are never made up.** Every number comes from deterministic *tools* that read the datasets; the AI model only chooses tools and writes the explanation.
- **Transparent.** Every answer shows the data date ("Data as of"), the source and a method panel explaining how the numbers were computed.
- **Usable without chat.** Every Dex also has feature tabs (charts, rankings, comparisons) built on the same tools.
- **Everywhere.** Web, an installable PWA that works offline, and Android and iOS apps. Available in Indonesian and English.

Who is it for?

Students, researchers, journalists, analysts and anyone who wants to understand market and economic data quickly without processing spreadsheets themselves.

> FinanDex provides information and education. FinanDex is **not** investment, financial or legal advice.

CHAPTER 02

The Problem

Financial and economic data is scattered across many datasets with different formats, terms and update dates. To answer a simple question such as "how did gold compare with Bitcoin over the last five years?" people usually have to find the data, clean it and compute the answer themselves.

Three main obstacles

1. **Fragmentation.** Stock, currency, macro, startup, cost-of-living and legal data live in separate sources.
2. **Technical barrier.** Computing returns, volatility, drawdowns or correlations requires data-analysis skills.
3. **AI hallucination risk.** General chatbots can state numbers that sound convincing but have no source.

The FinanDex approach

FinanDex separates two jobs: **computation** is done by tested deterministic code, while **language** is handled by the AI model. The result is an answer that is easy to understand, traceable and consistent — the same question on the same data always yields the same numbers.

CHAPTER 03

How It Works

Every question follows the same path, from your message to an answer with charts.

The path of a question

1. **You ask** in Finandex AI or directly in a specific Dex.
2. **Safety checks** on the device and on the server block messages containing sensitive data (seed phrases, private keys, card numbers, national ID numbers).
3. **The AI model picks tools.** The model receives that Dex's tools with strict schemas and decides which ones to call.
4. **Tools compute.** A tool reads the dataset and returns three things: compact data for the model, a chart/table specification for the screen, and a method note.
5. **The answer is written.** The model explains the tool results. Charts are chosen by the tool, not by the model.
6. **You see the result** streamed: text, interactive charts, tables, a method panel and source citations.

Deliberate limits

- At most 8 turns and 12 tool calls per question; each tool has an 8-second time limit.
- When data is unavailable, the Dex says so clearly ("pending data") instead of guessing.
- Out-of-scope questions are routed to a better-suited Dex.

Feature tabs without chat

Every Dex has tabs (for example Overview, Rankings, Compare, Data) that run the same tools directly in the browser. The Data tab always shows the source, license and data date.

CHAPTER 04

27 Dex in 8 Categories

FinanDex is built from a single Dex *registry*. Navigation, Home, the Dex Hub, search, the chat router and the data-sources page are all generated from it, so the list below always matches the app.

The Dex list

Category	Dex	Scope	Status
Global Markets & Equities	Finandex AI	Global market assistant: indices, commodities, crypto, currencies (main chatbot & router)	live
Global Markets & Equities	NASDAQDex	500 NASDAQ stocks, 1962–2026	pending data
Global Markets & Equities	SnP500Dex	233 S&P 500 stocks: total return, dividends, volume	pending data
Global Markets & Equities	CommerceDex	43 global e-commerce giants: prices, fundamentals, macro	live
Global Markets & Equities	IPODex	Global IPOs 2019–2026: India, US, UK & 100+ countries	live
Indonesian Markets & Economy	IDFinanDex	Indonesia multi-asset & macro: IHSG, USD/IDR, BI rate	live
Indonesian Markets & Economy	RupiahDex	Daily USD/IDR exchange rate since 2001	live
Indonesian Markets & Economy	IDXDex	Indonesia Stock Exchange equities: prices, fundamentals, IPOs	live
Indonesian Markets & Economy	ReksaDex	Indonesian mutual funds: AUM, investment managers, performance	live
Crypto & Global Macro	CryptDex	Crypto: prices, market snapshot, DeFi, protocol fees, bot education	live
Crypto & Global Macro	CentralBankDex	Central-bank balance sheets & global liquidity (Fed, ECB, BoJ, 167 countries)	live
Crypto & Global Macro	InflationDex	Inflation & cost of living across countries	live
Startups, VC & Companies	StartupDex	Global startup funding rounds 2019–2026	live
Startups, VC & Companies	VCDex	Startup investor directory 2026	live
Startups, VC & Companies	VCNetworkDex	Co-investment network between VC investors	live

Category	Dex	Scope	Status
Startups, VC & Companies	OpenCompanyDex	Directory of 100k companies (Open Company Database sample)	live
Cost of Living, Work & Household	SalaryDex	Global salaries by skill & role (100k-posting sample)	live
Cost of Living, Work & Household	CostLivingDex	Cost & quality of life in 148 cities (synthetic data)	live
Cost of Living, Work & Household	IDHousingDex	Indonesian subsidised & commercial housing (project listings)	live
Cost of Living, Work & Household	IDFoodDeliveryDex	Price benchmark of 10k Indonesian food-delivery menu items	live
Social & Development	PovertyDex	Multidimensional Poverty Index (MPI) for 108 countries	live
Social & Development	PopulationDex	Global population news coverage 2019–2026	live
Social & Development	HomelessnessDex	Global news coverage of homelessness 2019–2026	live
Social & Development	HousingDex	Global housing news coverage 2019–2026	live
Sentiment & Personal Finance	FintechDex	Fintech product adoption & reviews from 14 platforms	live
Sentiment & Personal Finance	PersonalFinanDex	Personal finance & financial independence discussions	live
Law	My LAIwyer	Legal information: the 1945 Constitution, old & new Criminal Procedure Code, Law 1/1946	partial

Finandex AI as the front door

Finandex AI answers cross-market questions (indices, commodities, currencies, crypto) and can open another Dex when your question is better answered there.

CHAPTER 05

Data & Methodology

FinanDex uses 42 input dataset files, processed into ready-to-use data by an ETL (*extract, transform, load*) step at build time.

Data principles

- **No invented numbers.** When a dataset is missing, the Dex is marked *pending data* and shows no substitute numbers.
- **Explicit data dates.** Every Dex shows "Data as of" taken from the dataset, not today's date.
- **Honest labels.** Synthetic data (for example CostLivingDex) and sampled data are clearly labelled in the interface.
- **Datasets are treated as data.** Dataset content is never executed as instructions or scripts.

Verifiable computation

- **Golden tests.** Results are compared with reference files (relative tolerance 1e-6, absolute 1e-9).
- **Python ↔ TypeScript parity.** Reference computations are independently re-implemented in Python; the results must match the app's implementation.
- **Method panel.** Every answer includes the formulas and assumptions used, such as how annualised return or drawdown is computed.

Data licenses

Each dataset has its own license, shown on the Data Sources page. CC BY-NC 4.0 datasets are for non-commercial use only; datasets with no stated license are flagged and need confirmation before public distribution.

CHAPTER 06

The AI Layer

FinanDex uses Anthropic's Claude models through the FinanDex server. The API key is never sent to the user's device.

How AI is used

- **Tool use.** The model can only call the active Dex's tools, with inputs validated against strict schemas.
- **Per-Dex prompts.** Each Dex has its own system prompt (ID and EN) on top of the same global rules.
- **Streaming.** Answers are sent incrementally (Server-Sent Events) so text and charts appear immediately.
- **Efficiency.** Prompts and tool lists are cached; a daily token budget caps cost.

Behaviour rules

- No buy/sell recommendations or personal legal advice.
- Always states the data date and source.
- Refuses to process sensitive data and reminds users not to share it.
- Admits when data is insufficient instead of guessing.

CHAPTER 07

Technical Architecture

FinanDex consists of a client app and one lightweight server function.

Client

- React 19, TypeScript (strict), Vite and Tailwind.
- ECharts loaded on demand; virtualised tables for large data.
- A PWA with a service worker: pages and core data are available offline.
- Capacitor for Android and iOS apps from the same codebase.

Server

- One standard Web handler (Request → Response) that runs as both a Vercel Function and a Netlify Function.
- A chat orchestrator: calls Claude, runs tools and streams results over SSE.
- The same deterministic tools run on the server (chat) and in the browser (feature tabs).

Data

Core data (`core`) ships with the app; heavy data (`heavy`, such as full price series and legal documents) is loaded on demand with caching.

CHAPTER 08

Security & Privacy

FinanDex is designed to need no account and to store as little data as possible.

Privacy

- No account, no ads, no third-party analytics or tracking.
- Chat history, watchlist and preferences are stored only on your device and can be deleted in Settings.
- The server logs only technical metadata (request id, Dex, tool names, latency, token counts) without message content.
- The privacy policy references Indonesian Law No. 27 of 2022 on Personal Data Protection.

Security

- Strict input validation and a 64 KB request size limit.
- An allowed-origin list (CORS), security headers and a Content Security Policy without inline scripts.
- Rate limits: 20 requests per 10 minutes and 200 per day per device, plus a daily token budget.
- Sensitive-data guards on client and server.
- Automated checks for leaked secrets in the repository and build output.

CHAPTER 09

Platforms & Accessibility

Available on

- **Web** — works in modern browsers.
- **PWA** — installable to the home screen and works offline for data already stored.
- **Android & iOS** — native apps built with Capacitor.

Built for everyone

- Indonesian and English.
- Light and dark themes, plus a colour-blind-friendly palette.
- WCAG 2.2 AA target: automated contrast checks, keyboard navigation and screen-reader labels.
- Table export to CSV with a source watermark.

CHAPTER 10

Quality Assurance

Every build must pass a series of automated gates. The results below were measured on the 21 October 2026 build.

Test results

Gate	Result
Unit + tool tests	380/380 passed
Coverage (statement / branch)	98.64% / 90.87%
Golden tests	60/60 passed, 49 files identical
Python ↔ TypeScript parity	48 files, 0 differences
Chat evaluation (mock, 128 items)	115 passed, 0 failed, 13 pending data
Playwright E2E (Chromium desktop + mobile)	116 passed
axe accessibility (8 pages × 2 themes)	0 serious violations
Lighthouse mobile, Home	Performance 91 · Accessibility 100 · Best Practices 100 · SEO 92
Initial JavaScript size	130.27 KB gzip (limit 250 KB)

Not yet measured

Chat evaluation with the live model (needs an API key), E2E on WebKit and Firefox, and the Android/iOS builds run in CI.

CHAPTER 11

Status & Limitations

Current status

- 24 Dex are live.
- NASDAQDex and SnP500Dex are **pending data** (the NASDAQ and S&P 500 datasets are not yet available).
- My LAIwyer is **partial**: the 1945 Constitution and Law 1/1946 are indexed; the Criminal Procedure Code (Law 8/1981) and the new Criminal Code (Law 20/2025) await their documents.

Limitations

- Data is historical as of each dataset's date; FinanDex is not a real-time price source.
- Some datasets are samples or synthetic, and are labelled in the interface.
- Maps are shown as scatter/bar charts because boundary data (GeoJSON) is not part of the inputs.
- The AI can make mistakes when explaining; always check the numbers in the method panel and the Data tab.

CHAPTER 12

Disclaimer & Copyright

Disclaimer

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Creator

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